

Derivatives: Power, Product, and Quotient Rules

Answer Key

AP Calculus AB/BC Differentiation

Quick Answers

1. $20x^4 - 6x$

2. $-\frac{10}{x^3} + \frac{3}{\sqrt{x}}$

3. $3x^2$

4. $6x^2 - 10x + 6$

5. $\frac{13}{(x+4)^2}$

6. $-x^2 \sin(x) + 2x \cos(x)$

7. $-\frac{4x}{(x^2-1)^2}$

8. $3\sqrt{x} + \frac{1}{2\sqrt{x}}, 6.25$

9. $\frac{2x^2-6x}{(2x-3)^2}, -4$

10. $\frac{-x \sin(x) + (x+1)(x \cos(x) + \sin(x))}{(x+1)^2}$

Curriculum

Level: AP Calculus AB/BC Differentiation

FUN-3.A–FUN-3.C – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 12 tagged checks.

1. $f(x) = 4x^5 - 3x^2 + 7$

Step 1. Differentiate term by term with the power rule $\frac{d}{dx}x^n = nx^{n-1}$: the exponent multiplies out front and drops by one.

Step 2. $4x^5 \rightarrow 20x^4$, $-3x^2 \rightarrow -6x$, and the constant 7 has slope zero.

$f'(x) = 20x^4 - 6x$

2. $g(x) = 6\sqrt{x} + \frac{5}{x^2}$

Step 1. The power rule needs powers, so rewrite: $g(x) = 6x^{1/2} + 5x^{-2}$.

Step 2. Differentiate: $6 \cdot \frac{1}{2}x^{-1/2} = 3x^{-1/2}$ and $5 \cdot (-2)x^{-3} = -10x^{-3}$.

Step 3. Rewrite with positive exponents.

$g'(x) = \frac{3}{\sqrt{x}} - \frac{10}{x^3}$

3. Limit definition for $f(x) = x^3$.

Step 1. Expand the binomial in the numerator: $(x + h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$, so the numerator is $3x^2h + 3xh^2 + h^3$.

Step 2. Every surviving term contains h , so factor it out and cancel — this is the step the definition is really testing:

$$\frac{h(3x^2 + 3xh + h^2)}{h} = 3x^2 + 3xh + h^2.$$

Step 3. Now the limit is a substitution: let $h \rightarrow 0$ and the last two terms vanish.

Agreement. The power rule gives $\frac{d}{dx}x^3 = 3x^2$, the same result — which is exactly why the rule is allowed to replace this work.

$$\boxed{f'(x) = 3x^2}$$

4. $y = (x^2 + 3)(2x - 5)$

Step 1. Product rule with $u = x^2 + 3$, $v = 2x - 5$, so $u' = 2x$ and $v' = 2$.

Step 2. $y' = u'v + uv' = 2x(2x - 5) + (x^2 + 3)(2)$.

Step 3. Expand and collect: $4x^2 - 10x + 2x^2 + 6$.

Check: expanding first gives $y = 2x^3 - 5x^2 + 6x - 15$, whose derivative is the same.

$$\boxed{y' = 6x^2 - 10x + 6}$$

5. $f(x) = \frac{3x - 1}{x + 4}$

Step 1. Quotient rule with $u = 3x - 1$, $v = x + 4$, so $u' = 3$ and $v' = 1$.

Step 2. $f'(x) = \frac{u'v - uv'}{v^2} = \frac{3(x + 4) - (3x - 1)}{(x + 4)^2}$.

Step 3. Simplify the numerator: $3x + 12 - 3x + 1 = 13$. The x -terms cancel, which is why this derivative is never zero.

$$\boxed{f'(x) = \frac{13}{(x + 4)^2}}$$

6. $h(x) = x^2 \cos x$

Step 1. Product rule with $u = x^2$, $v = \cos x$, so $u' = 2x$ and $v' = -\sin x$.

Step 2. $h'(x) = 2x \cos x + x^2(-\sin x)$. The minus sign comes from the derivative of cosine, not from the rule.

$$\boxed{h'(x) = 2x \cos x - x^2 \sin x}$$

7. $f(x) = \frac{x^2 + 1}{x^2 - 1}$

Step 1. Quotient rule with $u = x^2 + 1$, $v = x^2 - 1$, so $u' = v' = 2x$.

Step 2. $f'(x) = \frac{2x(x^2 - 1) - (x^2 + 1)(2x)}{(x^2 - 1)^2}$.

Step 3. The numerator is $2x^3 - 2x - 2x^3 - 2x$; the cubic terms cancel and $-2x - 2x$ remains.

$$\boxed{f'(x) = \frac{-4x}{(x^2 - 1)^2}}$$

8. $f(x) = (2x + 1)\sqrt{x}$

(a) **Step 1.** Product rule with $u = 2x + 1$ and $v = x^{1/2}$, so $u' = 2$ and $v' = \frac{1}{2}x^{-1/2}$.

Step 2. $f'(x) = 2\sqrt{x} + (2x + 1) \cdot \frac{1}{2\sqrt{x}}$.

Step 3. Split the second term: $\frac{2x}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} = \sqrt{x} + \frac{1}{2\sqrt{x}}$, so the two radical terms combine to $3\sqrt{x}$.

(b) Substitute $x = 4$: $3\sqrt{4} = 6$ and $\frac{1}{2\sqrt{4}} = \frac{1}{4}$, giving a tangent slope of 6.25.

$f'(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}, \quad f'(4) = 6.25$

9. $f(x) = \frac{x^2}{2x - 3}$

(a) **Step 1.** Quotient rule with $u = x^2$, $v = 2x - 3$, so $u' = 2x$ and $v' = 2$.

Step 2. $f'(x) = \frac{2x(2x - 3) - x^2(2)}{(2x - 3)^2}$.

Step 3. Numerator: $4x^2 - 6x - 2x^2$, which collapses to $2x^2 - 6x$.

(b) At $x = 1$ the numerator is $2 - 6 = -4$ and the denominator is $(2 - 3)^2 = 1$.

$f'(x) = \frac{2x^2 - 6x}{(2x - 3)^2}, \quad f'(1) = -4$
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10. $f(x) = \frac{x \sin x}{x + 1}$

Step 1. The outermost structure is a quotient, so start there: $u = x \sin x$ and $v = x + 1$, with $v' = 1$.

Step 2. u is itself a product, so the product rule gives $u' = \sin x + x \cos x$.

Step 3. Substitute into $\frac{u'v - uv'}{v^2}$.

$f'(x) = \frac{(\sin x + x \cos x)(x + 1) - x \sin x}{(x + 1)^2}$
